

CAPILLARY RISE METHOD (05)

Aim

To determine the surface tension of water using capillary rise method.

Materials and Apparatus

Five capillary tubes, travelling microscope, beaker, pointer, lab jack and stands.

Theory

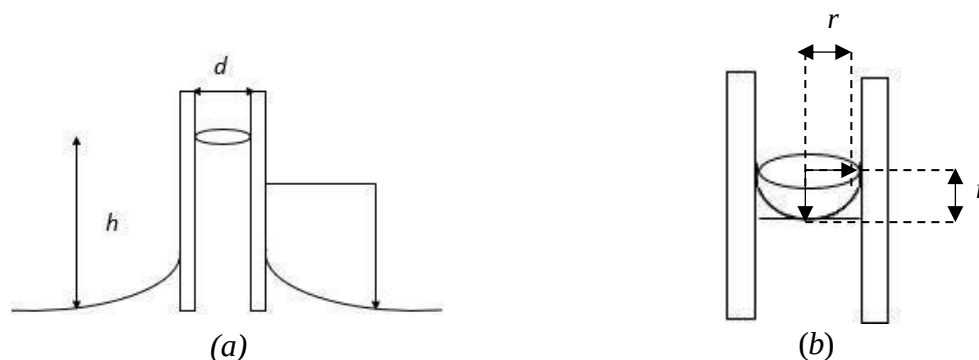


Fig. 1: (a) The setup of the experiment (b) Magnified water meniscus

Where, h : Height to which water rises in the capillary tube.

d : Internal diameter of the capillary tube, r : Internal radius of the capillary tube

For static equilibrium of the water column,

$$2\pi rT = mg \quad (1)$$

and

$$mg = \pi r^2 h \rho g + \left(\pi r^3 - \frac{2}{3} \pi r^3 \right) \rho g \quad (2)$$

where,

m : Mass of the water meniscus

T : Surface tension of water

ρ : Density of water

Above equation (1) and (2) results,

$$\frac{h}{r} = \left(\frac{2T}{\rho g} \right) \frac{1}{r^2} - \frac{1}{3} \quad (3)$$

Hence,

$$T = \frac{G\rho g}{2} \text{ where, } G \text{ is the gradient of the graph of } \frac{h}{r} \text{ vs. } \frac{1}{r^2}$$

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Procedure

1. **Q:** What should you do before using the given apparatus?
2. Arrange the setup suitably to do the experiment.
3. Measure the capillary rise using a capillary tube and repeat the procedure for other tubes.
4. **Q:** What are the other measurements required?